

HW4 Q4 Review — Relevant Lecture Material

From Slides09: Bernoulli Processes III — Law of Large Numbers

ORF309/MAT380

All content below is drawn from **ORF309_MAT380_S26_Slides09** (pp. 47–56). These are the concepts needed for Q4 parts (a) and (b).

Q4(a): Law of Large Numbers

Setup: The Sample Average

Given an original random variable X with $\mathbb{E}[X]$ defined, we repeat the experiment n times to get i.i.d. copies $X_1, X_2, \dots, X_n$. The **sample average** is:

$$\frac{1}{n} \sum_{k=1}^n X_k$$

This is itself a random variable for every finite n .

Expectation of the Sample Average (Slides09, p. 14)

$$\mathbb{E}\left[\frac{1}{n} \sum_{k=1}^n X_k\right] = \frac{1}{n} \sum_{k=1}^n \mathbb{E}[X_k] = \frac{1}{n} \cdot n \cdot \mathbb{E}[X] = \mathbb{E}[X]$$

The expectation of the sample average equals $\mathbb{E}[X]$ for *any* n (no limit needed).

Theorem: Law of Large Numbers (Slides09, p. 16–18)

Law of Large Numbers for Random Variables

With probability 1:

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n X_k = \mathbb{E}[X]$$

Key observations from slides:

- For every finite n , the sample average $\frac{1}{n} \sum_{k=1}^n X_k$ is **random**.
- In the limit $n \rightarrow \infty$, the sample average is **not random** — it equals $\mathbb{E}[X]$.

Think about how the Monte Carlo estimator $\hat{Z}_N = \frac{1}{N} \sum_{k=1}^N Z_k$ relates to this theorem.

Q4(b): Variance and the Proof of LLN

Definition of Variance (Slides09, p. 24)

$$\text{Var}(X) = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right]$$

A random variable becomes “less random” when it is closer to its expectation (i.e., the squared deviations shrink).

Key Variance Properties (Slides09, p. 25)

Variance Properties

1. $\text{Var}(X) \geq 0$
2. $\text{Var}(X) = 0$ if and only if X is not random
3. $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$
4. If $X \perp Y$, then $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$
5. $\text{Var}(\alpha X) = \alpha^2 \text{Var}(X)$, for all $\alpha \in \mathbb{R}$

Variance of the Sample Average (Slides09, pp. 27–31)

The proof of LLN computes the variance of S_n/n (the sample average). The slides work through this for indicator RVs, but the same logic applies to any i.i.d. random variables. The key computation pattern is:

$$\text{Var}\left(\frac{1}{n} \sum_{k=1}^n X_k\right) \xrightarrow{\text{Prop. 5}} \frac{1}{n^2} \text{Var}\left(\sum_{k=1}^n X_k\right) \xrightarrow{\text{Prop. 4}} \frac{1}{n^2} \sum_{k=1}^n \text{Var}(X_k)$$

Since the X_k are i.i.d. with $\text{Var}(X_k) = \text{Var}(X)$:

Variance of the Sample Average

$$\text{Var}\left(\frac{1}{n} \sum_{k=1}^n X_k\right) = \frac{1}{n^2} \cdot n \cdot \text{Var}(X) = \frac{\text{Var}(X)}{n}$$

Connection to Mean Square Error

Recall from the slides (p. 27) that the proof of LLN uses:

$$\mathbb{E}\left[\left(\frac{S_n}{n} - p\right)^2\right] = \text{Var}\left(\frac{S_n}{n}\right)$$

This works because $\mathbb{E}[S_n/n] = p$, so the squared deviation from the true mean *is* the variance. The same logic connects to the MSE definition in Q4(b):

$$\text{MSE}(N) = \mathbb{E}\left[(\mathbb{E}[Z] - \hat{Z}_N)^2\right]$$

Use the expectation result (sample average is unbiased) and the variance result above to compute $\text{MSE}(N)$ in terms of $\text{Var}(Z)$ and N .